

Base rule to compute partial derivatives $\leadsto$ Chain rule:

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- e depends on pre-activation $z_i^{(l)}$ through all the pre-activations $z_j^{(l+1)}$ in layer $(l+1)$ connected to the activation $a_i^{(l)}$
- let $N_i^{(l+1)} = \{ j \text{ in layer } (l+1) \mid z_j^{(l+1)} \text{ is connected to } a_i^{(l)} \}$

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Delta rule $\delta_i^{(l)} = f'^{(l)}(z_i^{(l)}) \sum_{j \in N_i^{(l+1)}} \delta_j^{(l+1)} \omega_{ji}^{(l+1)}$ $f^{(l)}$: layer (l) activ. func.